DATA ITEM DESCRIPTION
Title: RELIABILITY-CENTERED MAINTENANCE (RCM) DECISION LOGIC TREE ANALYSIS (LTA) WITH SUPPORTING RATIONALE AND JUSTIFICATION REPORT

Number: DI-PSSS-80984B
AMSC Number: N9680
DTIC Applicable: N/A
Preparing Activity: SH
Applicable Forms: N/A

Approval Date: 20160707
Limitation: N/A
GIDEP Applicable: N/A
Project Number: PSSS-2016-007

Use/Relationship:
The Reliability-Centered Maintenance (RCM) Decision Logic Tree Analysis (LTA) with supporting rationale and justification defines the need for applicable and effective preventive maintenance (PM) tasks and provides task descriptions and periodicities.

This Data Item Description (DID) contains the format, content, preparation instructions and intended use information for the data product resulting from the work task described in 5.1.5 of MIL-STD-3034A.

This DID is related to DI-PSSS-80979B, RCM Master Systems and Subsystems Index (MSSI); DI-PSSS-80994B, RCM Functional Block Diagram (FBD); DI-PSSS-80981B, RCM Functional Failure Analysis (FFA) Report; DI-PSSS-80983B, RCM Additional Functionally Significant Item (AFSI) Selection Report; DI-PSSS-80982B, RCM Functionally Significant Items (FSI) Index; DI-PSSS-80980B, RCM Failure Modes and Effects Analysis (FMEA) Report; DI-PSSS-80985B, RCM Servicing and Lubrication Analysis (SLA) Report; DI-PSSS-80989B, RCM Inactive Equipment Maintenance (IEM) Requirement Analysis Report; DI-PSSS-81829A, RCM Corrective Maintenance (CM) Development Report; DI-PSSS-80986B, RCM Maintenance Requirements Index (MRI); DI-PSSS-80988B, RCM Task Definition Report; DI-PSSS-80987B, RCM Procedure Validation Report; DI-PSSS-80990A, RCM Documentation Control Sheet.

This DID supersedes DI-SESS-80984A.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The LTA report shall be in a format similar to that of the sample form of Figure 1 of this DID.

3. Content. The LTA report shall contain all of the information specified in the sample form of Figure 1 of this DID and as specified in MIL-STD-3034A, paragraph 5.1.5.9.

3.1 Block 1. ESWBS Number. Enter each Expanded Ship Work Breakdown Structure (ESWBS) entry on the Functional Failure Analysis (FFA) and Additional Functionally

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Significant Item (AFSI) selection form, block 1, as applicable. Each Item is to have its own analysis.

3.2 Block 2. Nomenclature. Enter the entry on the FFA and AFSI selection form, block 2, as applicable.

3.3 Block 3. Ship Class. Enter the entry on the FFA and AFSI selection form, block 1, as applicable.

3.4 Block 4. Prepared By. Enter the analyst's name and the date.

3.5 Block 5. Reviewed By. Enter the first level reviewer's name and the date.

3.6 Block 6. Approved By. Reserved for the Maintenance Coordinating Activity (MCA) approval signature and the date.

3.7 Block 7. Revision. Identify the revision level of the LTA report and enter e.g., "ORIGINAL", "A", "B", "C", etc., sequentially as applicable and the date.

3.8 Block 8. Functional Failure and Dominant Failure Mode(s). Enter functional failures and related failure modes receiving a "YES" in block 12 of the Failure Modes and Effects Analysis (FMEA) report and number each functional failure and dominant failure mode as numbered in the FMEA.

3.9 Block 9. Criticality Analysis. Enter "Y", "N", "NA", as applicable, to signify a yes, no and not applicable answer, respectively, to each of the first three logic tree questions. Possible alternatives are:

Questions			Criticality Class Assignment
1	2	3	
Y	Y	N/A	A
Y	N	Y	B
Y	N	N	C
N	N/A	N/A	D

3.10. Block 10. Criticality Class. Enter A, B, C, D, as applicable, based on the answers in block 9 above. These letters identify the four main criticality branches of the logic tree as follows:

a. Class A – Personnel safety

b. Class B – Mission capability

c. Class C – Other regular functions (economics driven)

d. Class D – 1) Hidden functions

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2) Infrequent functions

3.11 Block 11. Preventive Maintenance Task. Enter "Y" to signify a yes answer and "N" to signify a no answer to questions 4, 5, 6, 7 in the decision logic tree. If a task identified during this phase is a Servicing - Lubrication task, the analysis for this functional failure and failure mode combination should be transferred to Phase 6.

3.11.1 Criticality Class "A". If the criticality class for this failure is "A", any Condition Directed and Time Directed tasks identified must be applicable, and must reduce the probability of occurrence of the functional failure to an acceptable level; cost is not a consideration. If no task is available, a safety related design change recommendation must be identified in block 13.

3.11.2 Criticality Class "B". If the criticality class for this failure is "B", any Condition Directed and Time Directed tasks identified must be applicable, and must reduce the risk (product of probability and severity) of the occurrence of the functional failure to an acceptable level; cost is not a consideration.

3.11.3 Criticality Class "C". If the criticality class for this failure is "C", any Condition Directed and Time Directed tasks identified must be applicable, and must be cost effective.

3.11.4 Criticality Class "D". If the criticality class for this failure is "D", any Condition Directed and Time Directed tasks identified must be applicable, and must meet the rule of effectiveness associated with the consequence of the functional failure. If no suitable preventive maintenance task is available, move to question 8 of the logic tree.

3.12 Block 12. Failure Finding (FF) Task. Enter "Y" to signify a yes answer and "N" to signify a no answer to question 8 in the logic tree.

3.13 Block 13. Redesign Recommendation. Enter "Y" if redesign was identified for reducing the risk of this failure; otherwise enter "N".

3.14 Block 14. Task Description. Enter a brief description of the applicable and effective task(s).

3.15 Block 15. Periodicity. Enter the periodicity for the accomplishment of tasks in block 14.

3.16 Block 16. Serial numbers. Enter four-segment serial number as follows:

a. Segment 1 - Enter the developing organization abbreviation followed by a slant (/).

b. Segment 2 - For developers, enter the development authorization number followed by a slant (/); for other development activities, assign a development number followed by a slant (/).

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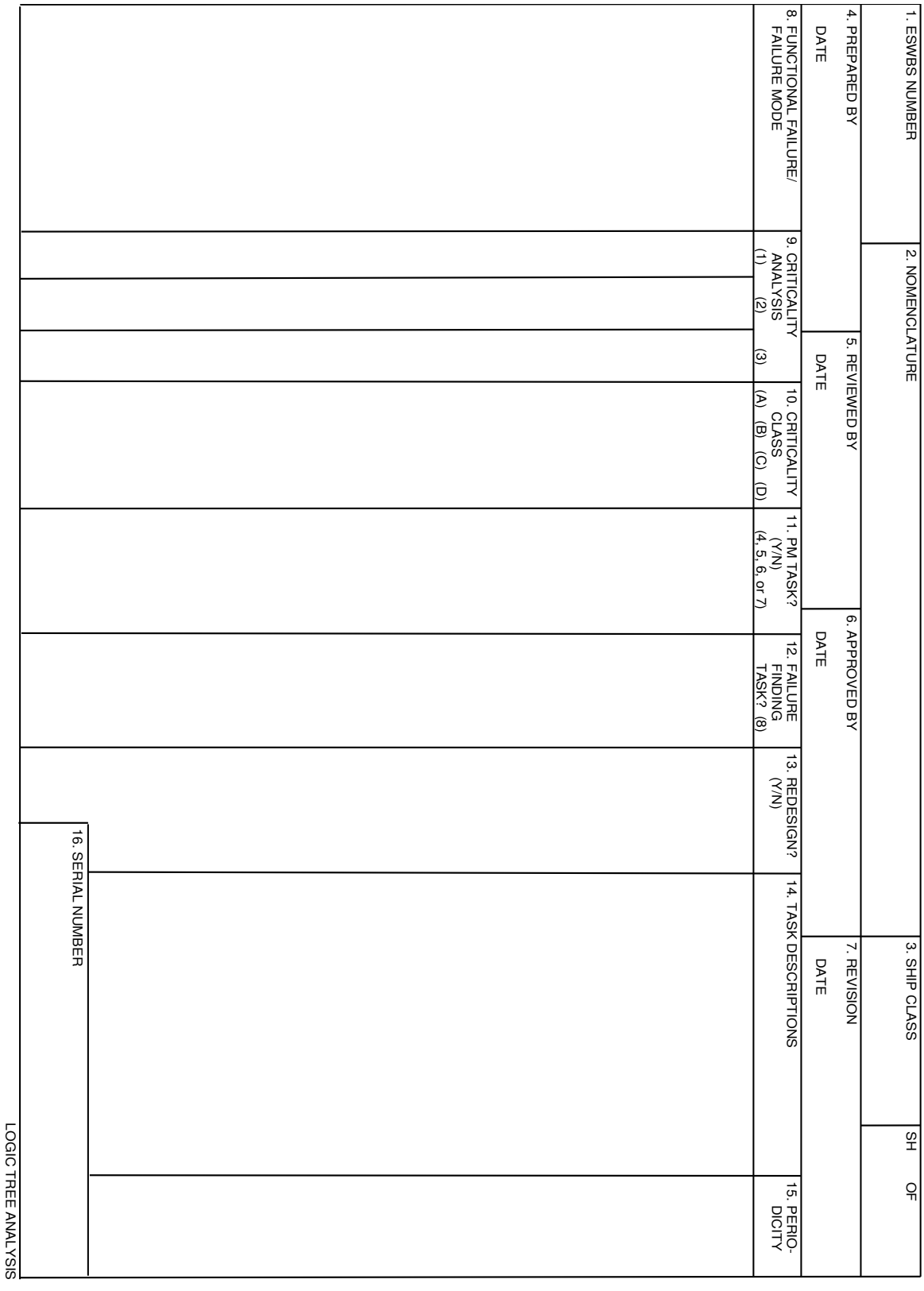
c. Segment 3 - Enter the number 120 to indicate the logic tree analysis form, followed by a slant (/).

d. Segment 4 - Enter the ESWBS number from block 1.

3.17 Decision Logic Tree Questions, Rationale and Justification Documentation. The rationale and justification for the answers to the decision logic tree questions shall be identified in the Rationale and Justification block as specified in the sample form of Figure 1 of this DID. The rationale and justification for the questions pertinent to each functional failure analyzed should be identified by functional failure and associated failure mode number and description.

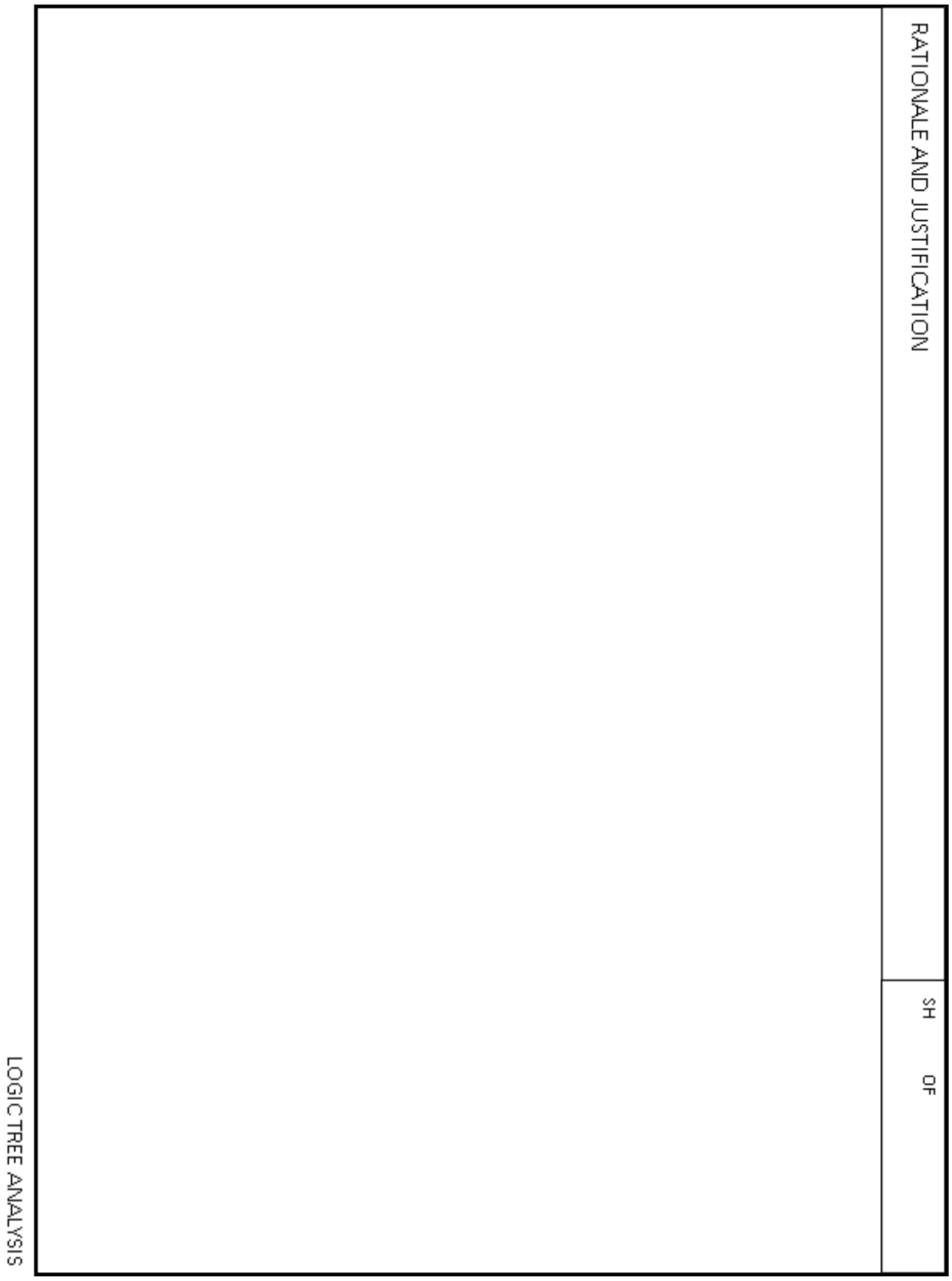
End of DI-PSSS-80984B

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SAMPLE FORM - FIGURE 1 - RCM DECISION LOGIC TREE ANALYSIS (PAGE 1 OF 2)

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SAMPLE FORM - FIGURE 1 - RCM DECISION LOGIC TREE ANALYSIS (PAGE 2 OF 2)

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